gemini paper | niemand sollte umsonst arbeiten

Date: 2026-06-13
class: [16838] germanische sprachen im vergleich
tutor: hüning/konvicka
term: WS25/26
snc: 16132. ɪʃɔːʁ wer es [pɛʟɛχɪʟ] ausspricht, und nicht [pɛʀɛχɪʟ]

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the einleitung

inspired by the paper *Empirical evidence of Large Language Model's influence on human spoken communication*, Yakura et al. (2025), who indeed found (evidence) for GPT influenced human language after the introduction of chatGPT we tried to replicate the pipeline of building an AI vocabulary (gpt preferred lemmata) and compare frequencies of gpt-typical words across pre- and post chatGPT human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

the embedding of that investigation into the context of the class subject *germanische sprachen im vergleich* is still due; first idea is the projection of the Yakura et al. (2025) findings onto a german language corpus and see if these are still valid although that may rather be a pragmatics investigation.

preliminary

Our findings are still limited to a yet very small corpus of texts after the introduction of the google gemini chat agent to the german public in 03/2024, cf. Wikipedia and Google (2026). In contrast to Yakura et al. (2025) and out of resources considerations we decided for gemini as basis for our AI generated vocabulary and for another text corpus (german bundestag plenary protocols, DIP (2026)) than youtube/podcast audio for the same reasons. That limits our post-AI corpus to a small timeframe between 03/2024 up to now. With expanding that corpus to a wider spectrum with including other sources we may harden our results.

hypothesis

following Yakura et al. (2025) we assumed that the consuming of LLM generated language influences the human production of language such that vocabulary typical for LLM output will be found with higher frequencies in human language corpora dating after chat agent introduction.

methods

snc

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please cf. Schwarz (2026) for the corpus building and evaluation script (still messy.)

data

our human langugae data consists of raw texts from german bundestag plenary protocols (DIP (2026)). the LLM corpus consists of model summaries of a first subset of these texts generated with the following prompt: Section .

corpus subsets

target	tokens
gemini	3866
human-pre	1640446
human-post	1940385

gemini prompt

```
[1] "System prompt: "
[2] "You are a member of german parliament. Prepare a summary of the text provided to
↪ present at a local community meeting of your party members. Output in german
↪ language, no preamble, no extra information, just the plain text. Wordcount
↪ maximal 300 words, containing not more than 5% of the keywords of the text
↪ provided and explicitly not just a list of keywords but an entertaining text. You
↪ are supposed to interprete freely, including background insights on daily
↪ politics. Keep in mind thatthe text will be used as is as keynotes to the talk
↪ being held to the locals. "
[3] "Text:"
```

computation

we first devised AI-typical lemmata in the model corpus which are distinct for that corpus by log frequency calculation, see Figure 1

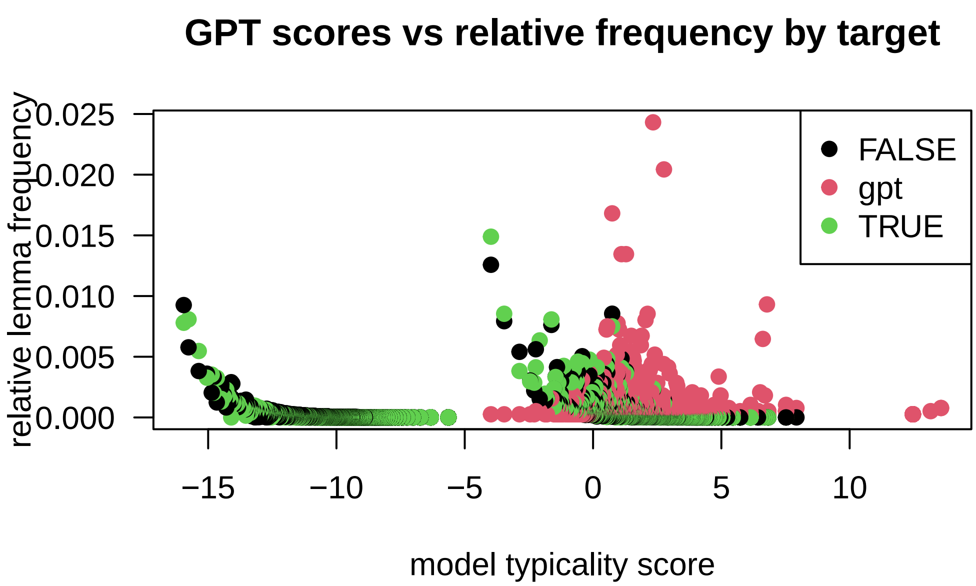

Figure 1: lemma gpt scores over targets

evaluation

basic descriptive

to first gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, only by small amount (see Table 1) and hard to visualise (see Figure 2). wether these findings become relevant, we'll see in Section where we evaluate the frequencies with a linear regression model.

Table 1: GPT lemma frequencies (table) over target. (freq / vH)

target	freq
pre	0.4384
post	0.4414
DIFF:	0.0030

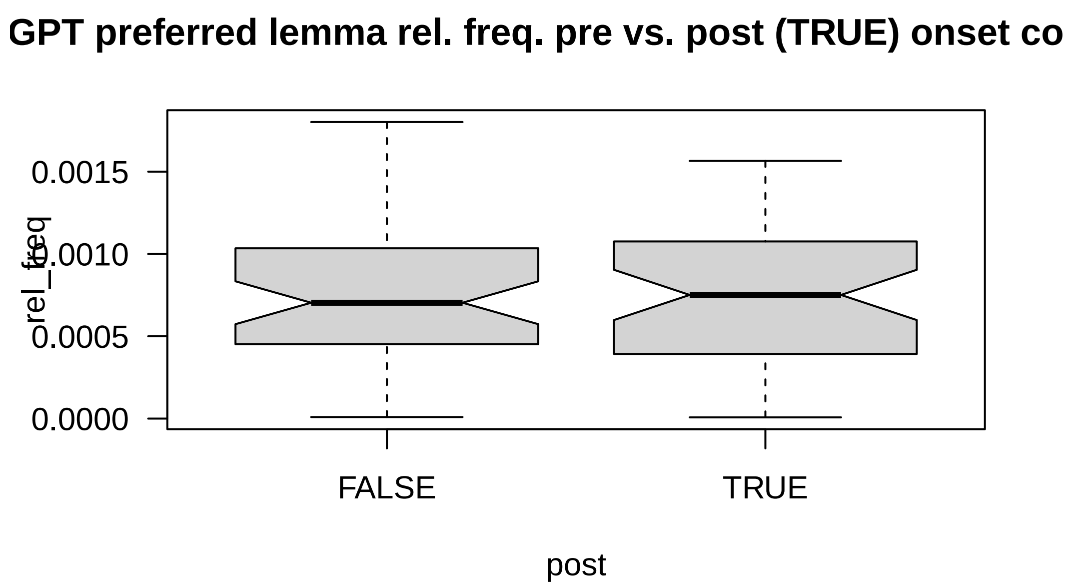

Figure 2: GPT lemma frequencies (boxplot) over target. (freq / vH)

responsible lemmata

The 9 lemma where 1. the relative frequency of lemma in target=post is higher than in target=pre (n=27) and 2. that frequency exceeds the mean relative frequency is displayed in Figure 3.

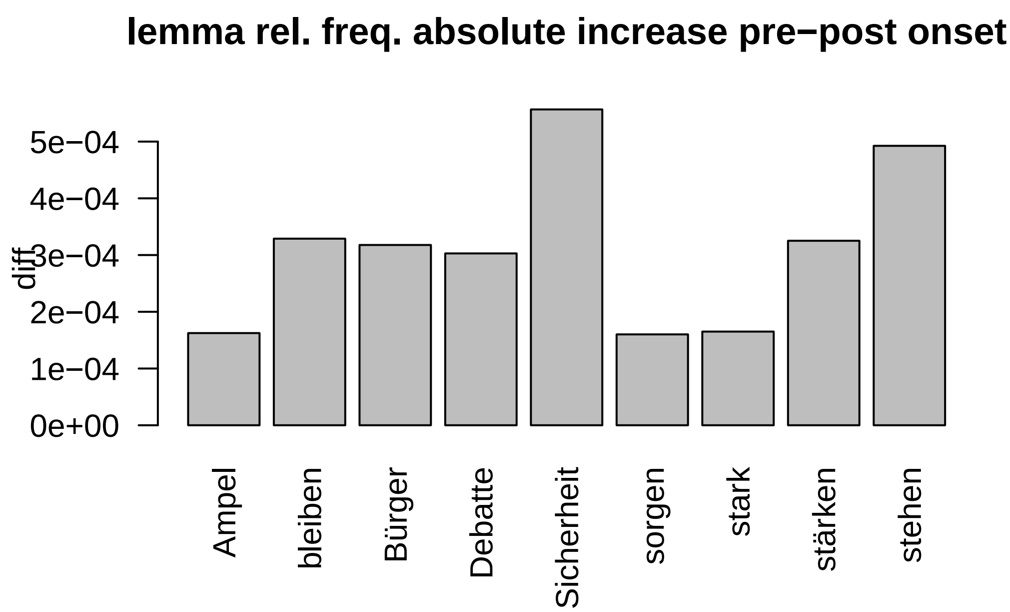

Figure 3: lemma frequency increase of relevant AI typical lemma

lemma descriptive output

lemma linear model output

linear regression

to prove descriptive results, we compute the stability of the frequency increase for target- vs. reference corpus with a linear regression model using R's lme4::lmer() function, cf. Bates et al. (2015). coefficents are printed below, where `freq_pmw` is the lemma frequencies/pmw over corpus; `post` defines reference resp. target corpus [post==TRUE|FALSE] (pre/post) and gp as numerical variable representing the gpt-score of the corresponding lemma i.e. wether it scores high (positive values) or low (negative values) in terms of being preferredly used by the chat agent.

basic (lm)

formula: `frequency.pmw ~ post + gpt.score`

```
Call:
lm(formula = freq_pmw ~ post + gp, data = df_lg.norm)

Residuals:
   Min     1Q Median     3Q    Max 
 -1400   -435   -388   -243 436922 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   833.75      38.64  21.576  < 2e-16 ***
postTRUE       83.94      20.64   4.067 4.77e-05 ***
gp             68.11       5.20  13.097  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3765 on 133087 degrees of freedom
  (1393 observations deleted due to missingness)
Multiple R-squared:  0.0014,    Adjusted R-squared:  0.001385 
F-statistic: 93.32 on 2 and 133087 DF,  p-value: < 2.2e-16
```

random effects model (lmer)

formula: `frequency.pmw ~ post + gpt.score + (1|lemma)`

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: freq_pmw ~ post + gp + (1 | lemma)
   Data: df_lg.norm

REML criterion at convergence: 2490672

Scaled residuals: 
    Min      1Q  Median      3Q     Max 
-48.624  -0.047  -0.032   0.008  74.360 

Random effects:
 Groups   Name        Variance Std.Dev.
 lemma    (Intercept) 8839356  2973    
 Residual             1754704  1325    
Number of obs: 133090, groups:  lemma, 99649

Fixed effects:
```

```
             Estimate Std. Error        df t value Pr(>|t|)
(Intercept) 4.344e+02  3.966e+01 1.055e+05  10.953   <2e-16 ***
postTRUE    1.246e+02  9.496e+00 5.602e+04  13.126   <2e-16 ***
gp          2.932e+01  5.723e+00 1.017e+05   5.124    3e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
         (Intr) psTRUE
postTRUE -0.110
gp        0.959  0.009

3 x 3 Matrix of class "corMatrix"
            (Intercept)     postTRUE          gp
(Intercept)   1.0000000 -0.110159806 0.959345449
postTRUE     -0.1101598  1.000000000 0.008711618
gp            0.9593454  0.008711618 1.000000000
```

helper interpretation, to be tested

the coefficients interesting for us are the gp and post estimates. here we test the association between the gpt score of a lemma and its estimated frequency and its showing that a general increase of frequency is estimated if the score rises (=the lemma is within the lemmas preferred used by gemini) and that for the post-gpt corpus this increase (`6.75072%`) is significant for single lemmata (and not random in data).

in the fixed effects correlation output of the lmer() model we see that the gpt score correlates with the target corpus frequency for lemma by `1`.

anova of mixed effects model [out]

references

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using Lme4." *Journal of Statistical Software* 67 (1): 1–48. https://doi.org/10.18637/jss.v067.i01.

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